

University AI infrastructure

Investment memorandum

DECISION REQUEST / INITIAL CONCEPT / CONSTANT USD ASSUMPTIONS

Recommendation

Defer an irreversible 25 MW construction commitment. Procure a cancellable, benchmarked lease pilot to measure demand. Preserve a conditional 6.25 MW facility option only after signed member commitments and binding grid terms. Massachusetts is an illustrative domestic candidate for an assumed US university consortium, not a proven global optimum.

Need and evidence

The assignment provides no signed long-term member commitments and no confirmed grid price or energization date. The 20 MW IT baseline, PUE 1.25 and full-year operation imply 25 MW and 219 GWh. An illustrative 14,000 GPU-equivalent fleet at 65% utilization, 99% availability and 90% job efficiency provides 71.0 million productive GPU-hours annually. This is modeled demand, not a survey or customer contract.

Three countries, with limits

The US EIA reports approximately 4,300 TWh of electricity generation in 2024 [S4]. China reports 10,086.88 TWh [S2], while Singapore reports approximately 60 TWh [S3]. National generation does not establish spare grid capacity. Singapore reports a 94% natural-gas fuel share in 2024 [S3]. The comparable World Bank renewable series returned data for 2015 but null for the queried 2022 period [S5]. It remains explicitly historical.

Ten-year comparison

Case / option	Before open \$ million	Annual opex \$ million	\$/productive GPU-hour	Capital risk \$ million
Base / Build	830.0	39.8	3.69	608.0
Base / Lease	59.8	239.1	3.37	59.8
Base / Hybrid	252.3	189.9	3.46	196.8
Delay / Build	1,091.5	39.8	4.00	608.0
Delay / Lease	59.8	239.1	3.37	59.8
Delay / Hybrid	497.1	189.9	3.53	196.8
Half use / Build	830.0	36.0	7.27	608.0
Half use / Lease	29.9	119.6	3.37	29.9
Half use / Hybrid	229.9	99.3	4.36	174.4

Calculated from labeled illustrative inputs, not market quotes. Annual opex is the first fully operating year. Delay means one extra year of leased bridge compute for owned options. All options serve equivalent demand. Capital risk is an assumed loss exposure, not a forecast loss probability.

What would justify ownership?

Model and financing boundaries

The model separates facility (\$250m), grid allowance (\$20m) and IT fleet (\$560m) at full scale. Facility cost is \$10m per total MW and IT fleet cost is \$40,000 per equivalent GPU. These inputs include conceptual power/cooling and non-GPU IT allocation, respectively. Electricity is \$0.09/kWh and leased allocation is \$3/GPU-hour. None is a verified quote.

The 10-calendar-year window begins at planned opening. Fleet replacements occur after four and eight operating years. Real discount rate is 8%, no tax or inflation is modeled, and residual value is zero. Debt funds 50% of facility/grid at 7% real interest with ten equal principal payments. Project unit cost excludes financing flows to avoid double-counting capital cost. Equity cash requirements display principal and interest separately.

Pre-opening cash is a gross funding requirement, including a three-month lease prepayment credited against later rent. Delay adds bridge operating costs, holding cost and interest. Capital risk assumes 80% loss on facility/grid, 70% on fleet, and three months of lease commitments. Idle energy remains 40% of full load before the variable utilization component. Changes to these assumptions may change the decision.

Engineering and risk allocation

For a conditional 6.25 MW phase, propose two independent 8 MVA transformer paths, N+1 UPS/cooling and five 2 MW generators with four required. At full phase load, a 48-hour outage requires 300 MWh. Assumed fuel use of 0.25 L/kWh plus 20% reserve implies 90,000 L. Verify derating, transfer tests, tank permits, fuel logistics and cooling continuity. No renewable or battery energy is credited as firm 48-hour supply.

A member nonprofit would own the shell and electrical/cooling assets and contract construction and qualified operations. Demand shortfall and member withdrawal require minimum-use commitments and reserves. Development equity carries early permitting loss. Construction debt requires permits, firm grid dates/costs, construction contingency and enforceable member contracts. Equipment finance requires delivery and acceptance tests, warranty and refresh terms.

Findings most likely to reverse the recommendation

- 1. Demand:** Signed commitments plus measured job traces demonstrate sustained utilization high enough to overcome ownership costs. Require synchronized-cluster benchmarks, not just annual GPU-hours.
- 2. Power:** A binding interconnection agreement fixes tariff, upgrade liability and date, with validated 48-hour resilience.
- 3. Comparable bids:** Delivered hardware and cloud bids show a durable ownership advantage after storage, networking, support, replacement and exit costs.

Governance and external effects

Use published quotas, metered variable charges and one institution-one vote. Reserve teaching access for smaller institutions, reclaim unused slots and offer appeals. Assess water allocation, noise, emissions, permits and whether grid upgrades shift costs to other customers. Data residency and hardware availability remain separate country-specific diligence items.

Source register

[S2] China NBS, 2024 communiqué. Accessed 2026-10-05. Full provenance in website Evidence.

[S3] Singapore EMA, Energy Transformation. Accessed 2026-10-05. Full provenance in website Evidence.

[S4] US EIA, 2024 energy statistics. Accessed 2026-10-05. Full provenance in website Evidence.

[S5] World Bank, EG.ELC.RNEW.ZS. Accessed 2026-10-05. Full provenance in website Evidence.